

Project Workflow: Week 1

Phase: Project Planning & Concept Finalization

1. Objective

2. The primary goal of Week 1 was to establish a robust foundation for a classic, interactive Ping Pong game delivering a smooth and engaging user experience. This included:
 - Defining the project scope and core gameplay boundaries.
 - Selecting an optimal Java-based technology stack for responsive real-time rendering and game logic.
 - Conceptualizing a clean, minimalist UI that keeps the player's focus on the game.

2. Detailed Task Breakdown

- **Feature Identification:** Conducted research on classic arcade Pong implementations to finalize the core feature set: smooth paddle and ball movement, responsive two-player controls, real-time score tracking, and a clean modern visual style.
- **Technical Stack Selection:** Selected Java with Object-Oriented Programming principles for clean state management, Swing for GUI rendering and the game canvas, the Timer class for the frame-rate-controlled game loop, and KeyListener for responsive, real-time paddle controls.
- **Architectural Setup:** Designed a modular, object-oriented class architecture separating the game loop, ball physics, paddle/collision logic, and score tracking to ensure clean code separation between rendering and game logic.

3. Deliverables

- **Project Scope Document:** A formal outline of the game's features and limitations.
- **Tech Stack Definition:** Finalized choice of Java, Swing, and the Timer-based game loop.
- **Project Timeline:** A structured 10-week development roadmap.

4. Tools & Resources

- **Development Environment:** Java JDK 11+ and IntelliJ IDEA / VS Code.
- **Reference Materials:** Java Swing, AWT, and Timer-based game loop references.